

PARDA: On-Device Audio Privacy for Smart Glasses using Small Language Models

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Abstract—Smart glasses can capture bystanders who have not consented to recording, exposing both their acoustic identity and private information conveyed through speech. Conventional PII redaction cannot reliably protect attributes inferred by linking seemingly innocuous statements across distant conversation turns. We present PARDA, an on-device audio privacy system that combines speaker-aware transcription, causal semantic anonymization, and non-source speech resynthesis. We extend SEAL to streaming multi-speaker conversation through bounded windows, persistent disclosure profiles, and retrieval of earlier evidence. Using 717 silver-labeled CANDOR conversations, we distill the adversary-anonymizer pipeline into a 2B small language model. On 143 held-out conversations, preference training reduces mean privacy leakage from 0.502 to 0.378 while preserving utility (0.875 to 0.900). The quantized Q4_0 model achieves 0.400 leakage and 0.902 utility. On a Raspberry Pi 5, the selected audio configuration achieves a speaker-mixture-weighted real-time factor of 0.7703. Across 40 concurrent end-to-end replay runs, mean transcription delay is 23.56 seconds, with a mean post-recording anonymization drain of 6.43 minutes for recordings averaging 31.07 minutes. These results characterize the privacy-utility and latency tradeoffs of causal audio anonymization on resource-constrained edge hardware.

I. INTRODUCTION

The growing popularity of smart glasses raises privacy concerns for bystanders who may be recorded without their consent. Figure 1 shows that modern devices resemble conventional eyewear and can therefore go unnoticed in everyday interactions [1], [2], [3], [4]. Existing privacy-enhancing systems for smart glasses focus mainly on images and video [5], [6], [7]. Audio introduces a different privacy problem. A speaker can be identified acoustically through characteristics of their voice, while the content of their speech can reveal identity, location, health, relationships, financial circumstances, and other private attributes.

Since private attributes can be inferred from contextual clues within and across conversation turns, protecting the transcript requires more than removing names, addresses, and other explicit identifiers. Figure 2 illustrates how an exchange about hospital night shifts can reveal a speaker’s former nursing role, while a recurring clinic schedule can suggest long-term hemodialysis without naming the treatment. Removing names and locations leaves this evidence intact: the disclosure lies not in an explicit identifier, but in what the remaining statements reveal together or imply. Prior work shows that language models can recover personal attributes from semantic cues that survive surface-level anonymization [8], [9]. The privacy problem is therefore not only identifying sensitive spans, but also recognizing when their context and relationships expose private information.

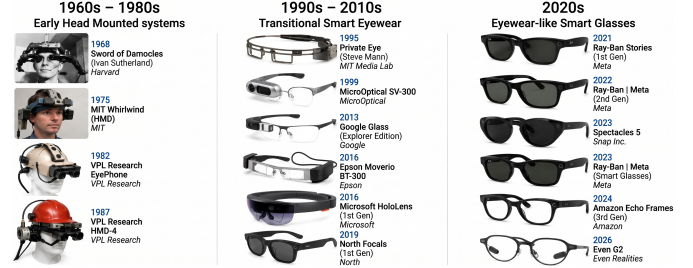


Fig. 1. Evolution of Smart Glasses throughout history.

Live smart-glasses audio makes this a causal decision problem, as a system producing an ongoing anonymized stream cannot wait for the full conversation and cannot retract information already obtained by an untrusted recipient. So the evidence needed to assess a disclosure may span the conversation, but the opportunity to intervene is confined to the current, unreleased window. Moreover, our threat model does not trust the companion phone with raw recordings, requiring anonymization inside the glasses trusted boundary. A practical system must therefore meet three design requirements: ① retaining relevant conversational history, ② selectively rewriting current speech while preserving useful meaning, and ③ operating within the compute and latency constraints of wearable hardware. These requirements motivate our central design question:

How can smart glasses protect both speaker identity and conversational content causally, without requiring the full conversation in advance?

We present PARDA¹ (**P**rivacy-preserving **A**udio **R**edaction with **D**ecryption on **A**uthorization), an on-device audio privacy system designed around these requirements². Its central design separates the history needed to assess a disclosure from the speech that can still be edited. A causal pipeline combining an adversary and an anonymizer maintains persistent per-speaker disclosure profiles, retrieves earlier evidence, and rewrites only the current window. The surrounding audio path separates and transcribes speakers and retains timing metadata for subsequent resynthesis in a non-source voice. The principal contribution is the streaming anonymization path. Consent-gated restoration is a supporting function adapted from prior work on reversible visual privacy [7].

This paper makes four contributions:

¹Parida is a Persian word meaning veil.

²Source code: <https://github.com/anonymoususer1356/anonymouspercom26>

- [C1] We address the dual privacy risks arising from *how speakers sound* and *what their conversations reveal* by coupling semantic anonymization with an interface for non-source voice resynthesis.
- [C2] We adapt SEAL [10] from static text to causal multi-speaker conversation, using bounded windows, persistent privacy state, and retrieval of earlier evidence to protect disclosures that span multiple windows.
- [C3] We construct privacy supervision from natural CANDOR conversations and distill the resulting adversary-anonymizer pipeline into a 2B model suitable for edge deployment, including quantized inference.
- [C4] We implement an edge prototype on Raspberry Pi 5 and characterize transcript-level privacy and utility, small-model inference cost, and multi-speaker audio scheduling, distinguishing the measured audio-path performance from the cost of language-model anonymization.

II. BACKGROUND AND RELATED WORK

A. Bystander Privacy

Smart-glasses privacy systems have primarily prevented capture, transformed visual content, or mediated consent for recorded media [5], [6]. Khawaja et al. support consent-mediated restoration of protected visual content [7], and other recent work consider bystander privacy in audio and audio-language systems as well [11]. PARDA addresses privacy in the audio channel, where risk can accumulate across a conversation rather than arise within a single frame.

B. Linked and Linkable Disclosures

We distinguish information stated explicitly from information that becomes private when evidence is combined across turns. Figure 2 illustrates this distinction from the perspective of an adversary: direct identifiers are visible in individual utterances, while other private attributes are recovered only by combining evidence across turns.

Linked identifiers, such as names and phone numbers, appear as localized tokens that conventional named-entity recognition can detect. *Linkable identifiers* acquire identifying force through the relation between pieces of evidence, even when each piece is ordinary on its own. In Figure 2, for example, a historical reference and school age together reveal an approximate age.

We measure this distinction using the number of lines cited as evidence and the *signal distance* between the first and last cited line (Section V-C0a). These are proxies, not definitions: evidence spanning multiple lines is not necessarily an inference. Table I lists the protected categories; each can contain linked or linkable information. The categories draw on data-protection instruments and indirect-identifier taxonomies [12], [13], [14], [15].

C. Text Anonymization and Attribute Inference

Conventional text de-identification targets explicit spans: rules, regular expressions, and named-entity models can identify names, addresses, organizations, dates, and account

TABLE I
PRIVACY TAXONOMY AND SUPPORTING SOURCES

Category	Scope	Examples	[12]	[13]	[14]	[15]
Identity	Direct identity	name, account, contact	✓	✓	✓	✓
Location	Place and movement	home, route, workplace	✓	✓	✓	✓
Demographic	Group attributes	age, nationality, religion	✓	✓	×	✓
Health	Physical and mental state	condition, medication	✓	✓	✓	×
Legal	Legal standing	visa, custody, litigation	×	×	×	✓
Economic	Financial position	salary, debt, employer	✓	✓	×	✓
Relational	Ties to others	partner, family conflict	×	×	×	✓
Lifestyle	Habits and routine	drinking, dating, hobbies	×	✓	×	✓
Appearance	Physical form	height, tattoo, disability	✓	✓	✓	✓

Marks record whether each instrument names the category among its enumerated protected classes. No single instrument covers all nine, so we protect their union.

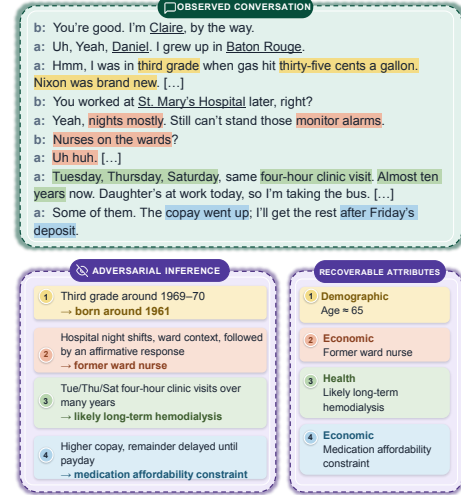


Fig. 2. Privacy leakage from linking conversational evidence across turns.

identifiers. These methods are well suited to locally expressed information, but cannot capture every risk that arises when otherwise ordinary statements are combined. Work on indirect and quasi-identifiers likewise shows that non-unique attributes can become identifying together, even when none is sufficient alone [16], [15], [17].

LLMs make this residual risk more consequential. Staab et al. show that they can infer location, demographic characteristics, and economic information from semantic cues that do not explicitly state those attributes [8]. Patsakis and Lykousas show that conventionally anonymized text can remain vulnerable to LLM-assisted re-identification [9]. Removing recognized entities therefore does not eliminate recoverable private information.

Inference can also guide protection. Staab et al. use an adversary to infer private attributes and an anonymizer to remove the supporting evidence, improving the privacy–utility tradeoff over entity-oriented anonymization [18]. Dou et al. generalize context-sensitive self-disclosures while retaining useful meaning [19], and RUPTA optimizes against LLM-based re-identification while preserving utility [20]. These approaches motivate language-model-based anonymization because the attribute being protected may not correspond to any single entity or span.

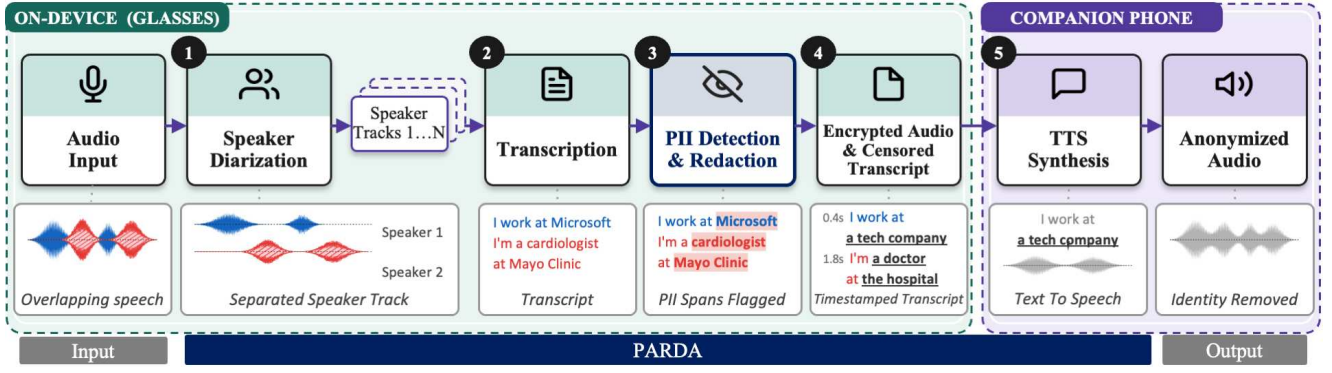


Fig. 3. PARDA processing path and trust boundary. Stages 1–4 execute within the glasses, while stage 5 runs on the companion phone.

D. SEAL

SEAL trains a small anonymizer through iterative adversarial interaction: an adversary infers private attributes from a fixed text, an anonymizer removes supporting evidence, and privacy and utility judgments guide distillation [10] for supervised fine-tuning and policy optimization. SEAL demonstrates that strong adversarial anonymization behaviour can be transferred into substantially smaller models suitable for edge deployment. Its static-document setting assumes that the full input is available and that successive candidates rewrite the same text.

III. THREAT MODEL

Live conversation requires a causal version of this process. PARDA uses speaker-addressed windows and retains relevant history through a disclosure profile and retrieval memory. It commits one edit per window, compares preference candidates from the same streaming state, and trains on line-grounded labels from natural conversation rather than synthetic persona labels.

We consider an adversary whose objective is to recover, in whole or in part, the identity or private attributes of a party whose speech was captured without that party having granted consent. A determined attacker can trivially defeat any capture-time protection by recording with an unrestricted device. We therefore scope protection to recordings made within the trusted-glasses boundary. Side channels, direct firmware compromise of the glasses, and wearer–cloud collusion are out of scope.

The wearer controls both the smart glasses and the companion smartphone. They may jailbreak the phone or the companion application in order to extract source-audio ciphertext retained for restoration, the resynthesized waveform, or the redacted transcript, and attempt to reconstruct the original speech or re-identify the bystanders present in it. We assume that the wearer cannot compromise the glasses’ trusted boundary, within which raw speech is processed and protected before release (Section IV). No plaintext source waveform leaves this boundary. This is an assumption about a production trusted computing boundary, rather than a security property enforced by our prototype. A deployed system would require secure boot, hardware-backed key storage, and

appropriate isolation of the trusted processing path. The phone may receive (i) source-audio ciphertext retained for consent-mediated restoration, whose decryption key is unavailable to the wearer without authorization; (ii) the redacted transcript and utterance-level timing metadata used for resynthesis; and (iii) a coarse, source-derived voice-category label. We treat the last two outputs as information disclosed to the wearer and discuss their residual leakage in Section VII; the phone never receives plaintext source audio.

IV. PARDA: OUR SOLUTION

PARDA is our principal contribution: a causal privacy pipeline that protects both a speaker’s acoustic identity and private information revealed through conversation. Live operation creates an irreversible release boundary: once speech leaves the glasses, earlier content cannot be revised in light of later disclosures. PARDA therefore processes conversation causally, carrying privacy state across bounded windows while rewriting only content that has not yet been released. Figure 3 presents the architecture and numbers the five processing stages used throughout the remainder of the paper.

- 1 *Speaker diarization and separation*: The glasses identify active speakers and maintain a track for each, allowing subsequent stages to reason about disclosures on a per-speaker basis. Overlapped regions are separated only when multiple speakers are active, avoiding unnecessary separation during single-speaker speech.
- 2 *Transcription*: Each speaker track is converted into a stream of speaker-addressed transcript lines, providing the textual representation required for semantic privacy analysis. Timing metadata is retained so rewritten content can later be aligned with the original conversational sequence.
- 3 *Causal semantic anonymization*: To protect disclosures that may depend on earlier conversation without revisiting already released speech, a causal adversary–anonymizer examines each bounded transcript window together with a persistent per-speaker disclosure profile and retrieved prior evidence. It redacts explicit identifiers and rewrites evidence that would otherwise expose private attributes across turns, while modifying only the current unreleased window.

TABLE II
DIARIZATION MODEL SELECTION

Model	Causal	Linear	Flex.	spk.	DER ↓
pyannote Community-1	×	×	✓		20.20
EEND-M2F	×	×	✓		16.07
EEND-TA	×	×	×		14.49
S2SD	×	×	✓		17.12
Streaming Sortformer	×	×	×		18.97
LS-EEND	✓	✓	✓		19.61

Sources: [21], [22], [23], [24], [25], [26].

TABLE III
PROTOTYPE VS. XR/AR HARDWARE

Spec.	Pt 5	XR2 Gen 1	AR2 Gen 1
Node	16 nm	7 nm	4 nm
CPU	4 × A76	8 × Kryo 585	4 × Kryo
Clock	2.4 GHz	2.84 GHz	≥ 1.9 GHz
RAM	LPDDR4X	LPDDR4X	LPDDR4X
NPU	×	✓	✓
Accel	×	Hex. 698, HTA 3rd-gen Hex.	Hex. HMX
Compute	0.61 INT8 TOPS	~15 TOPS	2.5 × XR2

[27], [28], [29], [30].

④ *Timestamped anonymized output*: Structurally valid edits are applied to the current lines and paired with their timing metadata, forming the protected interface between on-glasses processing and downstream use. Only this anonymized derivative is released beyond the glasses’ trusted boundary; source audio retained for authorized restoration leaves the boundary only as ciphertext.

⑤ *Speech resynthesis*: Because semantic redaction alone does not hide acoustic speaker identity, the companion phone converts the redacted, timestamped transcript into a non-source voice to produce anonymized audio.

Authorized restoration follows a separate consent-mediated side path adapted from prior visual privacy work [7]. In this path, any retained source audio is encrypted inside the trusted boundary before leaving the glasses, and the TTP (Trusted Third Party) mediates the release of the corresponding decryption key. We treat this as a fixed supporting mechanism rather than part of the evaluated anonymization contribution.

V. IMPLEMENTATION

Our implementation realizes the five-stage processing path shown in Figure 3. We describe the components and runtime mechanisms used at each stage to support causal, streaming operations.

① *Speaker Diarization, Separation, and Association*

The audio front end uses non-overlapping five-second scheduling chunks; model state persists across chunks, and replay benchmarks deliver chunks at real-time pace to prevent access to future audio.

a) *Diarization*: We deploy the DIHARD III LS-EEND checkpoint as an ONNX model [26]. LS-EEND provides causal frame-level activity, persistent long-form state, flexible speaker count, and linear temporal complexity. Although EEND-M2F and EEND-TA report lower offline DIHARD III DER, LS-EEND better matches the streaming edge setting (Table II). We use an activity threshold of 0.4 and support up to three simultaneously active speakers.

b) *Overlap separation*: LS-EEND activity triggers separation only during overlapping speech. Two- and three-speaker regions use the corresponding fixed-shape Sandglassset ONNX checkpoints [31], selected for their quality–size tradeoff (Figure 5c). Single-speaker regions bypass separation. The shared runtime assigns two intra-op threads to Sandglassset, and separated channels proceed independently to speaker association and ASR.

c) *Online speaker association*: ReDimNet-B1 [32] embeds only diarization-active speech. Each track maintains a sample-count-weighted, L2-normalized profile, and new channels are assigned by cosine similarity. Confident pairs are assigned greedily; unresolved matches remain ambiguous. The shared policy assigns three intra-op threads to ReDimNet-B1. Because the encoder is nonlinear, this profile is an approximate *online profile aggregation* rather than an exact embedding of the concatenated track.

d) *Post-recording fingerprint*: After recording, diarization-active regions from each completed track are batch-encoded with ReDimNet2-B6 [33] and combined into one weighted speaker fingerprint. This stage runs concurrently with the remaining SLM drain and is excluded from live audio-path RTF. The fingerprint is encrypted under a per-fingerprint symmetric key, which is wrapped using the TTP’s RSA public key.

② *Transcription*

We deploy Moonshine v2 Medium as the ASR backend. As shown in Figure 5e, it lies on the accuracy–size Pareto frontier among the evaluated speech models and provides the best operating point for our deployment, achieving 6.65% average WER. A single recognizer is shared across speakers, with each diarized speaker track transcribed independently.

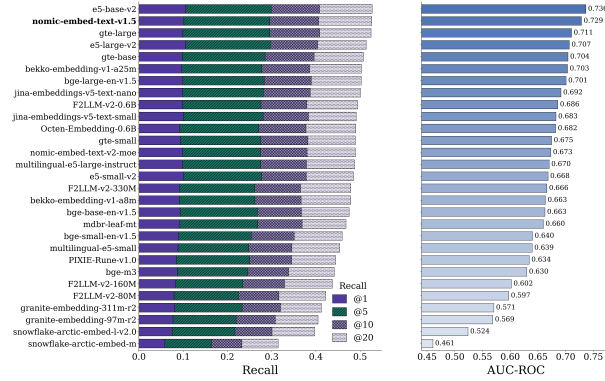
Turns are not cut at scheduling-chunk boundaries. Transcript state persists across chunks, and a turn is committed only when endpointing fires or the stream finalizes, subject to trailing-silence and maximum-utterance bounds. This preserves linguistic context and produces sentence-like turns that better match the units used to train the SLM. Partial hypotheses are kept separate from committed turns.

③ *Semantic Anonymization*

a) *Dataset and privacy labels*: We use CANDOR [34], a corpus of naturalistic conversations between previously unacquainted participants, to study privacy disclosures distributed across turns. Our processed pool contains 1,656 transcripts. Although unscripted and unconstrained in topic, these conversations are not representative of wearer-view audio.

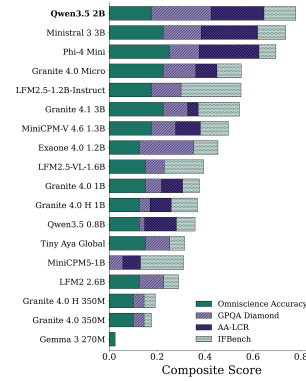
We linearise overlapping speech using CANDOR’s cliffhanger turn representation, which segments speech at floor changes rather than fixed silence thresholds. An audit of 200 transcripts (66,600 lines) found sentence-final punctuation in 99.9% of lines, supporting their use as sentence-like redaction units. We preserve fillers, disfluencies, and ASR artifacts to reflect deployment transcripts.

Since CANDOR lacks privacy labels, DeepSeek V4 Pro (XHigh Reasoning) extracts per-speaker attributes from complete transcripts under the nine categories in Table I. Each silver-labeled item contains an inferred value, confidence, and supporting lines for redaction supervision and signal-distance analysis. Two annotators subsequently verify the generated items. We label 717 transcripts (43.30%): 515 for training, 59 for checkpoint validation, and 143 for held-out evaluation.

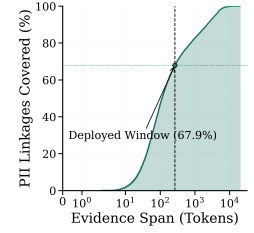


(a) Retrieval recall

(b) Separability



(c) SLM capability



(d) Linkage coverage

Fig. 4. Component selection and privacy-evidence coverage.

b) Streaming windows and state: We pack complete speaker-addressed transcript lines into an approximately 264-token window (12 CANDOR cliffhanger lines on average), preserving line references for edit directives. The budget covers 67.9% of observed privacy-evidence linkages (Figure 4d); the remaining 32.1% motivate persistent disclosure profiles and historical retrieval. The budget is selected through the privacy, utility, and lag search described below.

At window t , the model receives transcript lines w_t and causal state $s_t = (P_t, M_t)$, where P_t is the accumulated per-speaker disclosure profile and M_t stores prior line embeddings and accepted edits. The adversary identifies private attributes and updates the profile after validation:

$$z_t \sim \pi_\theta(\cdot \mid f_{\text{adv}}(P_t, w_t)), \quad P_{t+1} = P_t \oplus \text{valid}(z_t). \quad (1)$$

For each current line, the system retrieves up to k relevant historical entries above similarity threshold γ . Per-line retrieval prevents evidence for one utterance from displacing evidence needed by another. The anonymizer then uses the updated profile and retrieved history to edit only the current window:

$$a_t \sim \pi_\theta(\cdot \mid f_{\text{anon}}(P_{t+1}, M_t, w_t)). \quad (2)$$

Validation gates enforce structured adversary outputs, valid speaker and line references, and bounded edit replacements. Only validated disclosures and successfully applied edits enter persistent state. The append-only profile precedes the changing window in the prompt to enable prefix-equivalent KV-cache reuse. Previously released transcript lines are never regenerated.

c) Retrieval embedding selection: Semantic similarity alone need not identify lines that jointly support a private inference. We benchmark 29 embedding models on 29,368 positive evidence-line pairs and 20,871 gap-matched negatives from the labeled CANDOR pool. Pair separability is measured by AUC-ROC and within-conversation retrieval by recall@ k . nomic-embed-text-v1.5 [35] has the highest pair separability (0.7364; Figure 4b) and the second-highest mean retrieval score (0.3326; Figure 4a). We select it for its quality-size tradeoff among evaluated models (Figure 5a).

TABLE IV
HELD-OUT PRIVACY AND UTILITY

Model	Leakage ↓	Utility ↑
<i>Full precision</i>		
Luna teacher	0.355	0.886
Base	0.502	0.875
SFT	0.406	0.904
DPO	0.378	0.900
<i>Q4_0</i>		
Base	0.512	0.912
SFT	0.433	0.905
DPO	0.400	0.902

Student rows use Qwen3.5 2B.

TABLE V
ACOUSTIC PRIVACY AND RESYNTHESIS

Attacker	OO	OA	AA
ECAPA	0.75%	48.50%	50.25%
WeSpeaker	0.69%	50.75%	49.08%
ReDimNet2-B6	0.25%	50.25%	49.50%
WER (orig./anon.)	3.46/34.23%		
NISQA-TTS	3.73		
Utter. dur. err.	31.69%		
Span mid. MAE	1.63 s		
Span dur. err.	24.23%		

d) Small language model selection: We compare 18 non-reasoning models below 4B parameters using normalized scores for factual knowledge, GPQA Diamond, long-context reasoning, and instruction following. Before task training, small models do not reliably follow the full anonymization protocol, so these benchmarks serve as capability proxies rather than direct anonymization scores. Qwen3.5 2B has the highest composite score (0.778; Figure 4c) and lies on the size-quality Pareto frontier (Figure 5b).

e) Teacher configuration: We use Optuna’s GP sampler with logarithmic expected hypervolume improvement (LogEHVI) to jointly optimize window length, retrieval depth, similarity threshold, and refinement count for privacy leakage, utility, and modeled inference lag. Across 80 configurations evaluated on 10 development transcripts, the selected point uses $W \approx 12$ lines (264 tokens), $k = 8$, $\gamma = 0.774$, and one pass, achieving 0.238 leakage, 0.843 utility, and 15.6 minutes of modeled terminal lag.

We validate the selection by re-evaluating 30 configurations on each of two disjoint 10-transcript batches. The selected configuration remains in the top 20% under equal-weight normalized ranking, with Kendall’s $W = 0.779$ across all three batches (Figure 6).

GPT 5.6 Luna generates adversary and anonymizer completions without reasoning and anonymizer traces for distillation into the non-reasoning student [36]. DeepSeek V4 Flash judges privacy and utility.

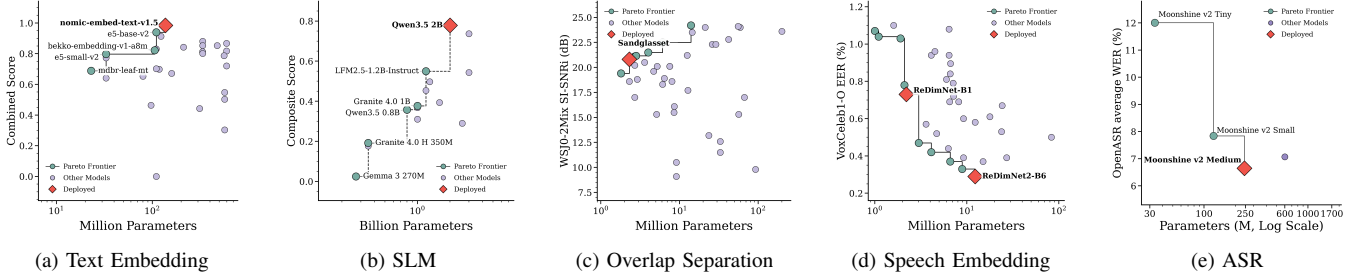


Fig. 5. Quality-size tradeoffs for component selection.

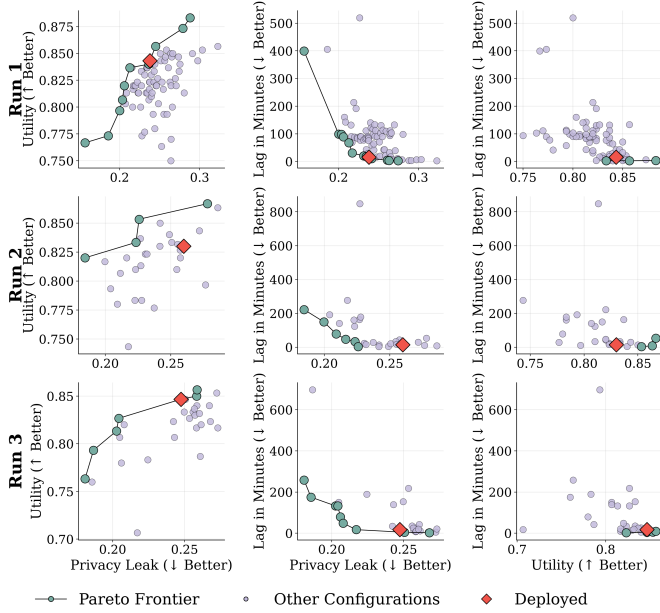


Fig. 6. Teacher-search transfer across samples.

f) Supervised fine-tuning: The 515 training conversations produce 15,480 windows and 30,960 balanced adversary and anonymizer examples. We optimize cross-entropy on teacher completions only, without sequence packing or judge outputs. The final step-3,870 adapter supplies the policy for preference collection and subsequent training.

g) Targeted preference optimization: Frozen-SFT diagnostics reveal persistent failures within privacy subtypes despite reasonable performance across the nine broad categories. We select 200 training conversations spanning high-, medium-, and low-leakage cases, covering 114 of 115 observed subtypes (99.1%). From each conversation, we sample five high-weakness and five complementary random windows for retention of what was learned during SFT.

For each window, we freeze the incoming state and sample eight anonymizer actions from the SFT policy (temperature 0.7, top- $p = 1$). DeepSeek V4 Flash evaluates each action against the same original window and silver-item bank, producing local leakage L and utility U (Figure 7). A precommitted action, independent of candidate rewards, advances the streaming trajectory. This isolates action-level preferences from differences

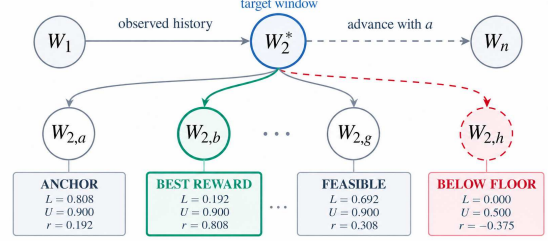


Fig. 7. Same-state preference construction. A precommitted action advances the trajectory independently of candidate rewards.

in conversation history.

Actions are ranked by

$$r(a) = \begin{cases} -1, & \text{structural failure,} \\ -(\tau - U)/\tau, & U < \tau, \\ 1 - L, & U \geq \tau, \end{cases} \quad \tau = 0.8. \quad (3)$$

The utility floor discourages privacy gains obtained through unusable rewrites. Groups require at least four distinct actions, a stable privacy item, and a reward spread of 0.1. After deduplication, pairs separated by a reward margin of at least 0.1 yield 6,241 preferences from 564 states across 190 conversations.

We apply sigmoid DPO against the frozen SFT reference [37], [10]. Rewards determine pair selection but do not weight the loss. Only anonymizer responses receive preference supervision; the resulting shared adapter serves both roles. We retain the one-epoch adapter. Training uses BF16 on one NVIDIA A100 80 GB and otherwise follows SEAL's hyperparameters, except for effective batch size.

h) Deployment, caching, and persistent state: For the base, SFT, and DPO variants, we merge any LoRA adapter into the BF16 backbone, export an f16 GGUF without the auxiliary MTP head, and apply llama.cpp Q4_0 weight quantization. Quantized inference uses llama.cpp with an f16 KV cache; nonquantized student comparisons use vLLM.

The live ONNX stages—LS-EEND, Sandglassset, ReDimNet-B1, and Moonshine V2 Medium—share full graph optimisation, sequential execution, one inter-op thread, memory-arena reuse, and disabled thread spinning. The runtime sweep varies per-stage intra-op thread budgets while holding model hashes and graph settings fixed. External OpenMP, MKL, and OpenBLAS pools are pinned to one thread to prevent competition with these budgets. ReDimNet2-B6 uses a separate

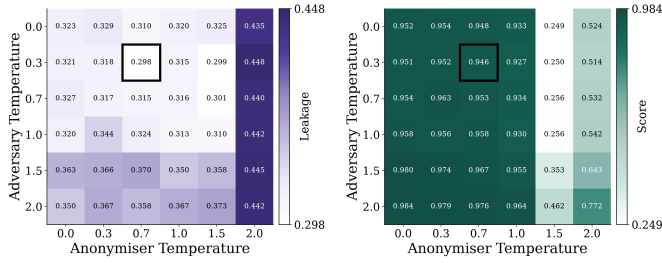


Fig. 8. DPO privacy leakage and utility across 36 adversary-anonymizer temperature pairs on 59 checkpoint-validation conversations. The outlined cell marks the selected operating point.

post-recording session and is excluded from the live scheduling sweep.

The SLM uses separate adversary and anonymizer slots so one role does not evict the other’s cached prefix. We reuse only prefix-equivalent prompt state: arbitrary KV blocks cannot be spliced across prompts because token positions and intervening causal context matter. Validated disclosure profiles, prior line embeddings, and applied edits persist across windows; rejected outputs do not.

i) *Training and deployment outcomes:* Table IV compares transcript-level leakage and utility for the teacher, nonquantized students, and Q4_0 students. SFT reduces mean leakage from 0.502 to 0.406 and increases utility from 0.875 to 0.904. Targeted DPO reduces leakage further to 0.378, near the teacher’s 0.355, with utility of 0.900. The paired SFT-minus-DPO leakage difference is 0.0276 (95% transcript-bootstrap interval: 0.0094–0.0453); the 0.0040 utility difference has an interval containing zero. These intervals capture conversation sampling, not training-seed variation.

Q4_0 preserves the improvement across training stages: leakage falls from 0.512 for the quantized base to 0.433 after SFT and 0.400 after DPO. Its DPO leakage exceeds that of nonquantized DPO (0.400 versus 0.378), while measured utility is similar. Because quantization also changes the inference backend, this compares deployment configurations rather than isolating weight-rounding effects.

j) *Inference-temperature sensitivity:* We evaluate the DPO model over 36 adversary-anonymizer temperature pairs on the 59 checkpoint-validation conversations. Figure 8 shows the resulting privacy-utility landscape. The lowest observed mean leakage is 0.298 at adversary/anonymizer temperatures of 0.3/0.7, with corresponding utility of 0.946. This setting also achieves the best combined score of 0.824, and is therefore used in the final configuration.

k) *Cross-window retrieval ablation:* We evaluate how much cross-window privacy evidence the final DPO model misses without retrieval. On the 59 checkpoint-validation conversations, disabling retrieval increases mean privacy leakage from 0.409 to 0.488 (Figure 9a), while mean utility decreases from 0.953 to 0.945 (Figure 9b). The no-retrieval variant also omits the retrieval instruction and stored-edit memory, so this comparison measures the contribution of the retrieval mechanism as deployed.

4 Timestamped Anonymized Output

The glasses transmit encrypted speaker tracks and the redacted transcript to the companion phone (Figure 3). To minimize disclosed timing information, the transcript retains only utterance start and end times, not word-level timestamps that could reveal the source speech’s temporal patterns. The TTS stage uses these coarse boundaries with jitter added to synthesize and align non-source speech without reproducing the original word-level timing, preventing unintentional disclosures.

5 Speech Resynthesis

The phone receives the redacted transcript, utterance-level timing spans, and a coarse voice-category label for non-source voice selection. An auxiliary L2-regularized logistic-regression classifier operates on frozen ReDimNet-B1 embeddings and achieves 98.2% macro-F1 on a speaker-disjoint VoxCeleb1 holdout. We use MAGIC-TTS because it exposes explicit duration and pause conditioning for timing-aware resynthesis.

On Table V, we evaluate acoustic privacy on LibriSpeech test-clean using 40 speakers, with five enrollment and ten disjoint trial utterances per speaker. We report equal-error rate (EER) for original-original (OO), original-anonymized (OA), and anonymized-anonymized (AA) verification under ECAPA-TDNN, WeSpeaker, and ReDimNet2-B6. Original speech is readily linkable at 0.25–0.75% EER, whereas OA verification approaches chance at 48.50–50.75%. ReDimNet2-B6, PARDA’s strongest post-recording speaker encoder, also yields 49.50% AA EER. These results show reduced source-speaker linkability under the three evaluated encoders.

This protection trades off against fidelity: WER rises from 3.46% to 34.23%, despite a NISQA-TTS score of 3.73. Mean utterance-duration error is 31.69%, with 24.23% relative span-duration error and 1.63 s span-midpoint MAE. These deviations indicate that source timing is not reproduced verbatim and may reduce exposure of temporal speaking patterns. Furthermore, Bounded timing perturbation could further harden this channel.

VI. EVALUATION

We ask two questions: whether PARDA protects privacy (Sec. VI-A), and what overhead the system incurs (Sec. VI-B).

A. Does PARDA Achieve Privacy?

a) *Setup and metrics:* All main comparisons use 143 held-out CANDOR conversations with 5,083 silver reference attributes. For each anonymized transcript, a fresh GPT-5.6 Luna attacker infers attributes, and DeepSeek V4 Flash scores each reference attribute as 0 (removed), 0.5 (generalized), or 1 (retained). *Leakage* is the mean score over items, then over conversations (lower is better). *Utility* averages readability R , meaning preservation M , and binary no-hallucination H : $U = (R/10 + M/10 + H)/3$ (higher is better). The local DPO reward in Eq. (3) uses a different, confidence-weighted scoring. The DPO operating point is selected from a sweep over 36 adversary-anonymizer temperature pairs on 59 checkpoint-validation conversations (Fig. 8).

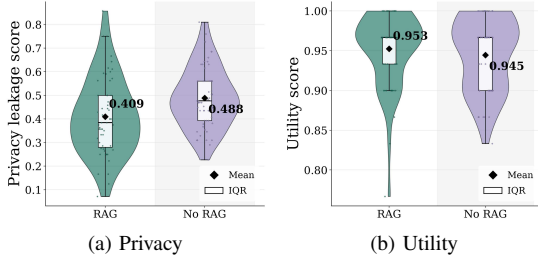


Fig. 9. Retrieval ablation on privacy and utility.

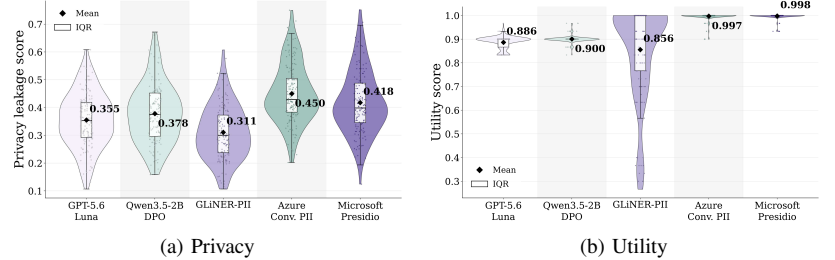


Fig. 10. PII baselines and full-precision DPO on 143 held-out conversations.

We compare the policy optimized Qwen3.5-2B SLM against the GPT-5.6 Luna teacher, and three conventional PII systems, GLiNER PII, Presidio, and Azure AI Conversational PII. The PII baselines replace detected line-level spans with category tags. Every arm uses the same transcript-level evaluation.

b) Results: Figure 10 compares the full-precision DPO model with the GPT-5.6 Luna teacher and conventional PII baselines. GLiNER achieves the lowest aggregate leakage at 0.311, followed by the teacher at 0.355 and PARDA at 0.378; Presidio and Azure reach 0.418 and 0.450, respectively. In utility, PARDA retains a score of 0.900, while GLiNER is lower at 0.856 and the more conservative Presidio and Azure baselines reach 0.998 and 0.997.

c) Comparison with conventional PII redaction: Although GLiNER achieves lower aggregate leakage, PARDA protects a substantially different subset of disclosures (Figure 13), uniquely protecting 10.7–15.5% of items missed by the PII baselines. Even relative to GLiNER, 10.7% of all labeled items are protected by PARDA while remaining exposed by the baseline. This behavior is also reflected across privacy categories: PARDA attains lower leakage than Presidio on most categories and lower than Azure on all but Identity, with its clearest advantage on Health (0.510 versus 0.548 for GPT-5.6 Luna and 0.616–0.626 for the PII baselines). Specialized PII detectors remain stronger for some explicitly expressed attributes, but the 2B SLM provides broader semantic coverage, handling both explicit and inferred disclosures without requiring a separate entity detector.

B. What Is the System Overhead?

a) Evaluation platform: We evaluate the on-device portion of PARDA on a Raspberry Pi 5 Model B [27]. The PARDA architecture itself is not tied to this platform; we use the Pi to characterize its computational behavior under resource-constrained edge compute. As Table III shows, the Pi provides four Cortex-A76 CPU cores but lacks the dedicated AI acceleration available on glasses-class platforms such as Snapdragon XR2 and AR2 Gen 1 [29], [30]. We therefore interpret the reported measurements as performance under a constrained CPU-only deployment rather than as direct estimates of production smart-glasses performance. We first characterize the cost of the live audio path across single- and multi-speaker workloads and determine a shared runtime configuration (Table VI). We then evaluate concurrent

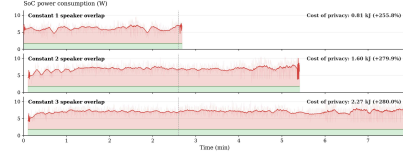


Fig. 11. Measured SoC power consumption under sustained one-, two-, and three-speaker workloads.

execution of the audio and semantic-anonymization pipelines using real-time replay on 20 CANDOR Audio files end to end with aggregate performance reported in Table VII.

b) Concurrent audio scheduling: We empirically sweep all thread allocations for the live audio pipeline on the four-core Raspberry Pi 5 and report real-time factor (the processing time divided by input audio duration).

Following the AMI Headset Mix distribution reported by Bullock et al. [38], we weight single-, two-, and three-speaker workloads by 81%, 15%, and 4%, respectively:

$$RTF_w = 0.81RTF_1 + 0.15RTF_2 + 0.04RTF_3. \quad (4)$$

Table VI reports the best measured allocation for each workload and the selected shared policy. The shared policy achieves a weighted audio-path RTF of 0.7703.

c) SLM inference cost: We measure prefill and decode throughput on the Raspberry Pi 5 across KV-cache depths of 0–3,000 tokens. Prefill decreases from 43.7 to 31.4 tokens/s, while decode decreases from 6.4 to 5.1 tokens/s. Although decode is costlier per token, prefill accounts for 66–71% of measured SLM processing time. These measurements yield a modeled SLM RTF of 1.32–1.39, separate from the measured audio-path RTF of 0.7703.

d) Concurrent end-to-end execution: We replay recorded conversations at real-time pace while the audio pipeline and SLM execute concurrently on the four-core Raspberry Pi 5. Figure 12 illustrates one execution, showing audio processing, adversary-anonymizer calls, and utterance-to-transcription delay. The audio-end marker separates processing during capture from the remaining post-recording drain.

Figure 11 shows the measured SoC power consumption under sustained speaker workloads, including the post-recording processing period.

We characterize end-to-end performance over 40 replay runs, reporting recording duration, transcription delay, and terminal anonymization lag (Table VII). Terminal lag is measured from

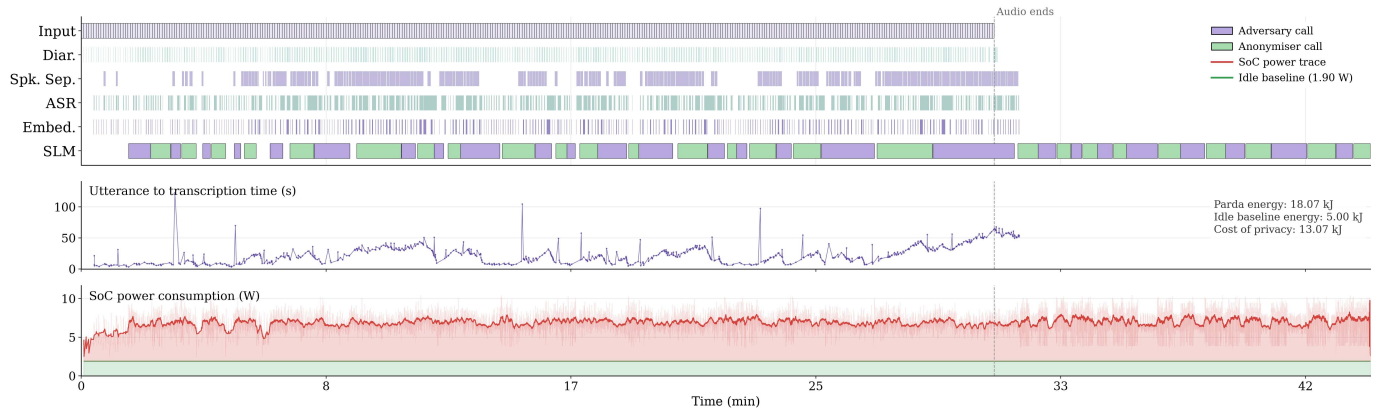


Fig. 12. Concurrent audio and SLM execution on Raspberry Pi 5. Top: per-stage execution timeline; middle: utterance-to-transcription delay; bottom: SoC power consumption.

TABLE VI
MEASURED AUDIO SCHEDULING

Workload	Diar.	Spk.	Emb.	OV.	ASR	RTF
1 spk.	1	–	4	–	3	0.6405
2 spk.	1	2	4	3	3	1.2471
3 spk.	1	2	1	3	3	1.5734
Weighted	1	2	3	2	3	0.7703

TABLE VII
MEASURED END-TO-END
PERFORMANCE

Metric	Result
Audio duration (mean, min)	31.07
Transcription delay (mean / P95, s)	23.56 / 52.92
SLM drain lag (mean / P95, min)	6.43 / 12.18
Maximum SLM drain lag (min)	12.82
PMIC power / energy (W / J)	6.94 / 17,725
Average privacy-stage overhead (%)	263

the end of audio capture to completion of the final anonymizer call. Power is sampled from the Pi 5 SOC PMIC.

Across the 40 runs, mean audio duration is XX minutes and mean terminal anonymization lag is XX minutes, with a 95th-percentile lag of XX minutes. These measurements capture contention between the audio front end and SLM, unlike the isolated scheduling and inference benchmarks.

VII. DISCUSSION AND LIMITATIONS

Speaker overlap and CPU contention can temporarily increase processing latency, but single-speaker speech dominates the evaluated workload and runs below real time. Consequently, periods of lower computational demand allow the pipeline to absorb earlier delays rather than letting every overlap-induced spike accumulate. Terminal anonymization lag depends on the remaining SLM workload when recording ends, particularly when privacy-relevant disclosures occur near the end of a conversation. Thus, transient processing delays need not translate into comparable terminal lag, although sustained overlap and outstanding SLM windows can extend processing beyond capture.

The privacy labels are model-generated and subsequently human-verified by two annotators. We label 717 of 1,656 CANDOR conversations, or 43.30%, rather than the complete corpus. CANDOR is English, dyadic, and recorded over video calls. It is closer to natural conversation than a structured meeting, but it is not an egocentric smart-glasses dataset.

The scheduler weighting is also not domain matched. The 81/15/4 split comes from four-person AMI meetings and conditions all percentages on voiced regions. Work on ordinary

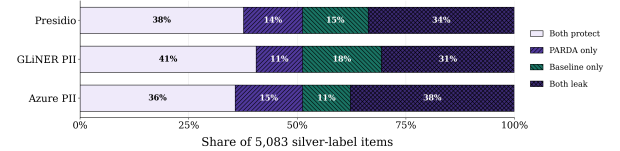


Fig. 13. Item-level complementarity between PARDA and conventional PII redaction.

turn-taking reports less than 5% simultaneous speech in the speech stream [39]. We use AMI as a reference workload, not a worst-case bound. The 4% three-or-more category is approximated by our three-source workload. Four-source costs and silence are not represented. A deployment study should derive this distribution from wearer-view audio and report conversation-level tails, not only a corpus average.

The Raspberry Pi 5 is a proxy rather than production smart-glasses silicon. The threat model also excludes compromise of the glasses and wearer-server collusion.

Restoration does not yet account for third-party disclosures. A segment may reveal PII about someone other than its speaker, and one speaker’s consent should not authorize restoring it. Flagging such segments as non-restorable is left to future work.

Resynthesized audio could carry a watermark marking it as synthetic. Because the wearer controls the phone where synthesis runs, they could strip it, so it is a mitigation rather than a guarantee. We leave it to future work.

VIII. CONCLUSION

PARDA addresses acoustic identity and semantic privacy in smart-glasses audio before it leaves the capture device. By combining causal anonymization, cross-turn retrieval, and a distilled language model, it reduces privacy leakage while preserving conversational utility. Our Raspberry Pi 5 prototype demonstrates the feasibility of on-device processing, with transient delays arising under speaker overlap and concurrent execution. Future work will evaluate domain-matched wearer-view audio, production smart-glasses hardware, and consent-aware restoration.

REFERENCES

- [1] M. Koelle *et al.*, “Your smart glasses camera bothers me: Exploring opt-in and opt-out gestures for privacy mediation,” in *NordiCHI*, 2018.
- [2] T. Chock, S. J. Kim, and Y. Lee, “Privacy concerns about ai smart glasses: A cross-cultural study of the u.s. and south korea,” in *Proceedings of the 59th Hawaii International Conference on System Sciences*, 01 2026.
- [3] A. Gallardo, C. Choy, J. Juneja, E. Bozkir, C. Cobb, L. Bauer, and L. Cranor, “Speculative privacy concerns about ar glasses data collection,” *Proceedings on Privacy Enhancing Technologies*, vol. 2023, pp. 416–435, 10 2023.
- [4] D. Bhardwaj *et al.*, “In focus, out of privacy: The wearer’s perspective on the privacy dilemma of camera glasses,” in *Proc. CHI*, 2024.
- [5] Aditya *et al.*, “I-Pic: A Platform for Privacy-Compliant Image Capture,” in *Proc. 14th Annu. Int. Conf. Mobile Syst., Appl., Serv. (MobiSys)*, 2016.
- [6] M. Corbett *et al.*, “BystanderAR: Protecting bystander visual data in augmented reality systems,” in *Proc. 21st Int. Conf. Mobile Systems, Applications, and Services (MobiSys)*, 2023, pp. 370–382.
- [7] Y. Khawaja, E. Nabeel, S. Humayun, E. Javed, K. Kromholz, M. H. Alizai, and N. Anwar Bhatti, “Now you see me, now you don’t: Consent-driven privacy for smart glasses,” in *2026 IEEE International Conference on Pervasive Computing and Communications (PerCom)*, 2026, pp. 1–10.
- [8] R. Staab, M. Vero, M. Balunovic, and M. Vechev, “Beyond memorization: Violating privacy via inference with large language models,” in *International Conference on Learning Representations*, B. Kim, Y. Yue, S. Chaudhuri, K. Fragkiadaki, M. Khan, and Y. Sun, Eds., vol. 2024, 2024, pp. 33 832–33 878. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2024/file/9028b8a3ca98f58e373f0c1497a17448-Paper-Conference.pdf
- [9] C. Patsakis and N. Lykousas, “Man vs the machine in the struggle for effective text anonymisation in the age of large language models,” *Scientific Reports*, vol. 13, p. 16026, 2023.
- [10] K. Kim, H. Jeon, and J. Shin, “Self-refining language model anonymizers via adversarial distillation,” in *Advances in Neural Information Processing Systems*, D. Belgrave, C. Zhang, H. Lin, R. Pascanu, P. Koniusz, M. Ghassemi, and N. Chen, Eds., vol. 38. Curran Associates, Inc., 2025, pp. 165 629–165 659. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2025/file/f1f0843df6f133309dd6d43d14f8a4e3-Paper-Conference.pdf
- [11] X. Zhan, G. Sun, J. Such, and P. Woodland, “Protecting bystander privacy via selective hearing in audio llms,” in *Annual Meeting of the Association for Computational Linguistics*, 2025. [Online]. Available: <https://api.semanticscholar.org/CorpusID:283896744>
- [12] European Parliament and Council of the European Union, “Regulation (EU) 2016/679 of the european parliament and of the council (general data protection regulation),” *Official Journal of the European Union*, L 119, 2016. [Online]. Available: <https://eur-lex.europa.eu/eli/reg/2016/679/oj>
- [13] California State Legislature, “California consumer privacy act of 2018,” *Cal. Civ. Code §1798.100 et seq.*, 2018. [Online]. Available: https://leginfo.ca.gov/faces/codes_displaySection.xhtml?lawCode=CIV§ionNum=1798.140
- [14] U.S. Department of Health and Human Services, “Standards for privacy of individually identifiable health information (HIPAA privacy rule),” 45 C.F.R. Parts 160 and 164, 2002. [Online]. Available: <https://www.hhs.gov/hipaa/for-professionals/privacy/index.html>
- [15] I. Baroud, L. Raithel, S. Möller, and R. Roller, “Beyond de-identification: A structured approach for defining and detecting indirect identifiers in medical texts,” 2025.
- [16] I. Pilán, P. Lison, L. Øvrelid, A. Papadopoulou, D. Sánchez, and M. Batet, “The text anonymization benchmark (TAB): A dedicated corpus and evaluation framework for text anonymization,” *Computational Linguistics*, vol. 48, no. 4, pp. 1053–1101, 2022.
- [17] M. Litvak and A. Jorge, “Recent advances in text anonymization: A systematic review,” *WIREs Data Mining and Knowledge Discovery*, vol. 16, no. 3, p. e70127, 2026.
- [18] R. Staab, M. Vero, M. Balunovic, and M. Vechev, “Language models are advanced anonymizers,” in *International Conference on Learning Representations*, Y. Yue, A. Garg, N. Peng, F. Sha, and R. Yu, Eds., vol. 2025, 2025, pp. 98 558–98 598. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2025/file/f478b2e8ad9ff0756bf5b79fb31c330f-Paper-Conference.pdf
- [19] Y. Dou, I. Krsek, T. Naous, A. Kabra, S. Das, A. Ritter, and W. Xu, “Reducing privacy risks in online self-disclosures with language models,” in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, L.-W. Ku, A. Martins, and V. Srikumar, Eds. Bangkok, Thailand: Association for Computational Linguistics, Aug. 2024, pp. 13 732–13 754. [Online]. Available: <https://aclanthology.org/2024.acl-long.741/>
- [20] T. Yang, X. Zhu, and I. Gurevych, “Robust utility-preserving text anonymization based on large language models,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Vienna, Austria: Association for Computational Linguistics, Jul. 2025, pp. 28 922–28 941. [Online]. Available: <https://aclanthology.org/2025.acl-long.1404/>
- [21] pyannoteAI, “Community-1: Unleashing open-source diarization,” pyannote Community-1 model documentation, 2025, diHARD III full-set DER benchmark.
- [22] M. Härkönen, S. J. Broughton, and L. Samarakoon, “EEND-M2F: Masked-attention mask transformers for speaker diarization,” in *Proc. Interspeech*, 2024, pp. 37–41.
- [23] S. J. Broughton and L. Samarakoon, “Pushing the limits of end-to-end diarization,” in *Proc. Interspeech*, 2025, pp. 5218–5222.
- [24] M. Cheng, Y. Lin, and M. Li, “Sequence-to-sequence neural diarization with automatic speaker detection and representation,” *IEEE Transactions on Audio, Speech and Language Processing*, 2025.
- [25] I. Medennikov, T. Park, W. Wang, H. Huang, K. Dhawan, J. Wang, J. Balam, and B. Ginsburg, “Streaming sortformer: Speaker cache-based online speaker diarization with arrival-time ordering,” in *Proc. Interspeech*, 2025, pp. 5238–5242.
- [26] D. Liang and X. Li, “LS-EEND: Long-form streaming end-to-end neural diarization with online attractor extraction,” *IEEE Transactions on Audio, Speech, and Language Processing*, vol. 33, pp. 3568–3581, 2025.
- [27] Raspberry Pi Ltd, *Raspberry Pi 5 Product Brief*, Raspberry Pi Ltd, 2026.
- [28] Arm Limited, *Arm Cortex-A76 Software Optimization Guide*, version 10.0 ed., Arm Limited, 2021.
- [29] Qualcomm Technologies, Inc., “Snapdragon xr2 5g platform,” Qualcomm Product Documentation, 2020.
- [30] Qualcomm Technologies, Inc., “Snapdragon ar2 gen 1 platform,” Qualcomm Product Documentation, 2022.
- [31] M. W. Y. Lam, J. Wang, D. Su, and D. Yu, “Sandglassnet: A light multi-granularity self-attentive network for time-domain speech separation,” in *2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2021, pp. 5759–5763.
- [32] I. Yakovlev, R. Makarov, A. Balykin, P. Malov, A. Okhotnikov, and N. Torgashov, “Reshape dimensions network for speaker recognition,” in *Proc. Interspeech*, 2024, pp. 3235–3239.
- [33] I. Yakovlev and A. Okhotnikov, “Redimnet2: Scaling speaker verification via time-pooled dimension reshaping,” *arXiv preprint arXiv:2603.11841*, 2026.
- [34] A. Reece, G. Cooney, P. Bull, C. Chung, B. Dawson, C. Fitzpatrick, T. Glazer, D. Knox, A. Liebscher, and S. Marin, “The candor corpus: Insights from a large multimodal dataset of naturalistic conversation,” *Science Advances*, vol. 9, no. 13, p. eadf3197, 2023. [Online]. Available: <https://www.science.org/doi/abs/10.1126/sciadv.adf3197>
- [35] Z. Nussbaum, J. X. Morris, B. Duderstadt, and A. Mulyar, “Nomic embed: Training a reproducible long context text embedder,” 2024.
- [36] Y. Li, X. Yue, Z. Xu, F. Jiang, L. Niu, B. Y. Lin, B. Ramasubramanian, and R. Poovendran, “Small models struggle to learn from strong reasoners,” 2025. [Online]. Available: <https://arxiv.org/abs/2502.12143>
- [37] R. Rafailov, A. Sharma, E. Mitchell, S. Ermon, C. D. Manning, and C. Finn, “Direct preference optimization: Your language model is secretly a reward model,” in *Advances in Neural Information Processing Systems*, vol. 36, 2023. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2023/hash/a85b405ed65c6477a4fe8302b5e06ce7-Abstract-Conference.html
- [38] L. Bullock, H. Bredin, and L. P. Garcia-Perera, “Overlap-aware diarization: Resegmentation using neural end-to-end overlapped speech detection,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2020, pp. 7114–7118.
- [39] S. C. Levinson and F. Torreira, “Timing in turn-taking and its implications for processing models of language,” *Frontiers in Psychology*, vol. 6, p. 731, 2015.